```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2); //Initialize
LCD
void setup() {
 Serial.begin(115200);
 lcd.init();
 lcd.backlight();
 lcd.clear();
 pinMode(2, OUTPUT); //Giving Pins
 digitalWrite(2, HIGH);
 delay(1000);lcd.setCursor(0, 0);
 lcd.print("IoT Idyllic PUMP"); //For Intro
 lcd.setCursor(0, 1);
 lcd.print("Activated ");
 lcd.print("");
 delay(5000);
 lcd.clear();
}
void loop() {
 int value = analogRead(A0);
Serial.println(value);
 if (value > 950) { //When soil is DRY
 digitalWrite(2, LOW);
```

```
 if (value > 950) { //When soil is DRY
 digitalWrite(2, LOW);
 lcd.setCursor(0, 0);
 lcd.print("Pump is ON");
 }
 else {
 digitalWrite(2, HIGH);
 lcd.setCursor(0, 0);
 lcd.print("Pump is OFF");}
 if (value < 500) {
 lcd.setCursor(0, 1);
 lcd.print("Moisture : HIGH");
 } else if (value > 500 && value < 950) { //
When soil is semi-dry
and semi-wet
 lcd.setCursor(0, 1);
 lcd.print("Moisture : MID ");
 } else if (value > 950) { //When Soil is wet
 lcd.setCursor(0, 1);
 lcd.print("Moisture : LOW ");
 }
}
```